Weekly Report – Week 3

Topic: Alert Triage & Correlation

Objective:

The objective of this week was to understand the concept of alert triage, differentiate between true positives and false positives, and perform correlation between alerts and system logs. The goal was to analyze alerts, verify them using sysmon logs, and determine whether the activity was real or simulated.

Activities Performed

1. Alert Triage

- Opened and analyzed **alerts_day1** dataset
- Reviewed alerts based on severity levels (Critical, High, Medium, Low)
- Understood different types of alerts **(A0020-A0030)** such as:
    - Credential Dumping
    - Command and Control (C2) Beacon
    - Malware Detection
    - Data Exfiltration
    - Unauthorized File Access
    - Brute Force Login Attempts

- Classified alerts as:
  - ✅ True Positive
  - ❌ False Positive
  - ❓ Needs Investigation

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2. Keyword Mapping

Each alert was converted into meaningful keywords based on attack behavior:

- Credential Dumping → mimikatz
- C2 Beacon → powershell, encoded, hidden
- Malware → .exe files
- File Access → cmd.exe, copy, type
- Data Exfiltration → copy, transfer
- Failed Logins → authentication attempts

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3. Log Correlation

Correlation was performed using sysmon_logs.csv by searching for relevant behaviors:

- Used Splunk queries such as:
  - `source="sysmon_logs.csv" mimikatz`
  - `source="sysmon_logs.csv" powershell`
  - `source="sysmon_logs.csv" cmd.exe`

- Focused on:
  - Process execution
  - Command line arguments
  - Suspicious flags such as -enc, -hidden, -nop
- Verified whether the activity mentioned in alerts actually exists in logs

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Key Findings

Suspicious Activities Detected

- PowerShell executed with encoded and hidden commands:
  - powershell -nop -w hidden -enc ...
  - Indicates Hidden Activity and attacker behavior
  - Splunk Query: source="sysmon_logs.csv" powershell
- Execution of credential dumping tool:
  - mimikatz.exe detected in logs
  - Confirms credential theft activity
  - Splunk Query: source="sysmon_logs.csv" mimikatz
- Suspicious command execution using cmd:
  - File copying operations observed
  - Splunk Query: source="sysmon_logs.csv" cmd.exe

Alerts Confirmed as True Positives

- Credential Dumping Tool Detected (e.g., execution of mimikatz.exe to extract passwords)
    - Splunk Query: source="sysmon_logs.csv" mimikatz
- Command and Control Beacon (with encoded PowerShell) (e.g., powershell -nop -w hidden -enc <encoded_command>)
    - Splunk Query: source="sysmon_logs.csv" powershell
- Malware Signature Match (e.g., detection of suspicious executable files like unknown.exe or mimikatz.exe)
    - Splunk Query: source="sysmon_logs.csv" "*.exe"
- Access to Restricted Files (e.g., cmd.exe /c copy sensitive_file.txt C:\temp)
    - Splunk Query: source="sysmon_logs.csv" cmd.exe

Alerts Requiring Investigation

- Data Exfiltration (partial evidence found)
  Example: File copy activity like cmd.exe /c copy mimikatz.exe C:\temp\ was observed, but no evidence of data being transferred outside the system.
    - Splunk Query: source="sysmon_logs.csv" copy
- Abnormal File Encryption Activity (no clear proof)
  Example: Multiple .exe executions were found, but no signs of encryption behavior such as mass file modification or ransomware patterns.
    - Splunk Query: source="sysmon_logs.csv" "*.exe"

- Multiple Failed Logins (not present in sysmon logs)
  Example: No authentication-related events were found in sysmon logs to confirm repeated login failures.
  - Splunk Query: Not applicable (no login data in sysmon logs)

Alerts Identified as False Positives

- Low severity login failures
  Example: Minor login attempts without repeated failures or suspicious patterns.
- Normal PowerShell activity without suspicious flags
  Example: PowerShell commands executed without -enc, -hidden, or other obfuscation flags, indicating normal administrative usage.
  - Splunk Query: source="sysmon_logs.csv" powershell

Correlation Methodology

The correlation process followed these steps:

1. Understand the alert type
2. Map the alert to attack behavior
3. Identify relevant keywords
4. Search for those keywords in sysmon logs
5. Analyze whether the activity is normal or suspicious
6. Classify the alert based on evidence

Learning Outcomes

During this week, the following skills were developed:

- Understanding of alert triage process
- Ability to differentiate between true and false positives
- Knowledge of attacker techniques such as:
    - Hidden PowerShell execution
    - Credential dumping using mimikatz
- Hands-on experience with log correlation
- Understanding that correlation is based on behavior, not exact field matching(IP,Usernames)

Recommendations

- Monitor PowerShell activity for encoded or hidden execution
- Block execution of known malicious tools like mimikatz
- Use SIEM tools for continuous monitoring

Conclusion

The Week 3 activities provided practical experience in alert triage and log correlation. By analyzing alerts and verifying them using sysmon logs, it was possible to identify real threats and seperate from false positives. This process helped in understanding how security analysts investigate incidents and make informed decisions based on evidence.